

Math 10A

Quiz 11; Tuesday, 8/7/2018

Time: 3 PM

Instructor: Roy Zhao

Name: _____

Circle True or False. (1 point for correct answer, 0 if incorrect)

1. True False If 1 is an eigenvalue of A , then there is a nonzero vector $\vec{v}$ such that $A\vec{v} = \vec{v}$.
2. True False If two matrices A, B have the same eigenvalues, then they have the same solutions to $\vec{y}' = A\vec{y}$.

Show your work and justify your answers. Please circle or box your final answer.

3. (10 points) (a) (8 points) Solve the system given by
$$\begin{cases} y_1'(t) = y_1(t) - 2y_2(t) \\ y_2'(t) = -4y_1(t) - y_2(t) \end{cases} \quad \text{and} \quad y_1(0) = 3, y_2(0) = 0.$$

- (b) (2 points) Find the least square error of using $y = x + 1$ as the line of best fit through the points $(0, 0), (1, 2), (2, 3)$.